

SOFTWARE REQUIREMENTS SPECIFICATION

GPunch — Secure On-Site Presence Verification System

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1. INTRODUCTION

1.1 Product Scope

GPunch is a secure, passwordless on-site presence verification system designed for institutions that require tamper-proof attendance records. The system replaces

paper registers, PIN-based check-ins, and selfie-based apps with a three-factor presence proof that requires no passwords, no maps, and no biometrics.

The product consists of two integrated components:

- **Backend REST API** — Node.js 18 + Express + MongoDB Atlas, handling user management, OTP authentication, geofence configuration, attendance records, and security audit logging.
- **Android Mobile Client** — Kotlin (API 26+), providing employee registration, authentication, GPS-based clock-in/clock-out, and background geofence monitoring.

In Scope

Feature	Description
Domain-restricted self-registration	Only institutional email addresses are accepted
Passwordless OTP authentication	6-digit OTP sent to registered email; no password required
Single-device binding	Each account is permanently tied to one Android device (ANDROID_ID)
GPS geofence clock-in / clock-out	Server-side Haversine validation; client cannot override
Mock GPS detection	Fake GPS apps detected and denied at the Android client
Automatic clock-out	Background service triggers clock-out when user leaves the geofence
Offline clock-out queueing	WorkManager retries the clock-out when connectivity is restored
Admin portal	Geofence setup, domain management, device reset, CSV export, audit logs

Out of Scope

- Web-based employee self-service portal
- Biometric authentication (fingerprint, face recognition)
- IoT beacon or NFC-based proximity attendance
- Integration with HR or payroll management systems
- Multi-campus / multi-geofence support (v1.0 is single-site)

1.2 Product Value

Traditional attendance systems suffer from well-known vulnerabilities:

Problem	Impact
Paper registers	Easily forged; no digital audit trail
PIN / password systems	Credentials shared between colleagues (buddy punching)
Selfie-based check-in	Privacy concerns; spoofable with photographs

Problem	Impact
Map-based check-in (client-side)	Location can be falsified by mock GPS apps

GPunch addresses all of these through three independently verified security pillars:

+	-----+	+	-----+	+	-----
-+					
	WHO are you?		WHICH device?		WHERE are you?
	Institutional		ANDROID_ID		GPS Geofence
	Email OTP		Device Binding		Haversine Check
	No password		One device only		Must be on campus
	Domain check		Per account		Server-validated
+	-----+	+	-----+	+	-----
-+					

Measurable outcomes:

- Zero buddy-punching: device binding makes it physically impossible to clock in for another person.
- Zero GPS spoofing impact: server re-validates coordinates independently of the client.
- Zero data loss on network failure: WorkManager ensures clock-out records are never lost.
- Full audit trail: every security event (domain rejection, OTP failure, device mismatch, mock GPS) is logged and visible to administrators.

1.3 Intended Audience

Reader	Purpose of This Document
Software Developers	Implementation reference for backend routes, models, and Android components
Faculty Guide / Evaluators	Academic assessment of scope, completeness, and design decisions
System Administrators	Deployment, geofence configuration, and day-to-day administration
Quality Assurance / Testers	Basis for writing, executing, and tracing test cases
Future Maintainers	Onboarding and change-impact analysis

1.4 Intended Use

Primary use case — Employee clock-in/clock-out:

1. Employee opens the GPunch Android app at the start of the workday.

2. The app verifies that high-accuracy GPS is enabled; if not, a system dialog prompts to enable it.
3. Employee taps **Clock In**. The app checks that the GPS location is not mocked, then sends coordinates to the server.
4. The server validates that the employee is within the configured geofence radius using the Haversine formula.
5. A successful clock-in starts a background foreground service that continuously monitors location.
6. When the employee leaves the geofence (or manually taps **Clock Out**), the attendance record is closed.
7. If the device is offline during auto clock-out, the payload is queued by WorkManager and submitted when connectivity is restored.

Secondary use case — Administrator management:

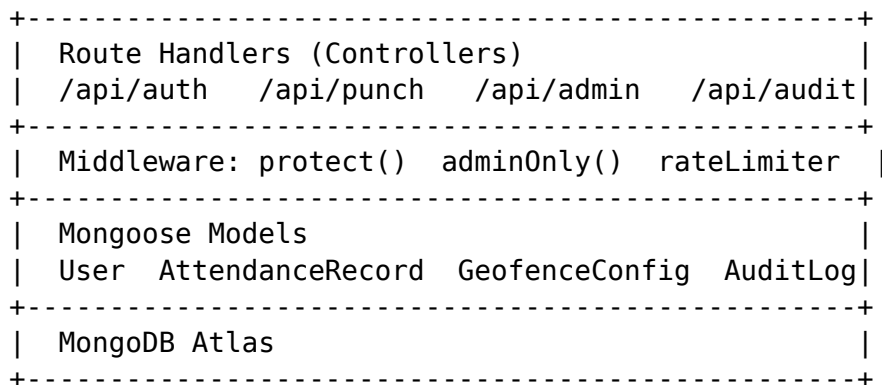
1. Admin calls POST /api/admin/geofence to set the campus centre coordinates, allowed radius, and institutional email domain.
2. Admin monitors GET /api/admin/audit-logs for security events (domain violations, OTP failures, device mismatches, mock GPS).
3. Admin calls POST /api/admin/reset-device/:userId to unlink a user's device (e.g., lost phone).
4. Admin downloads GET /api/admin/export-csv for payroll or HR reporting.

1.5 General Description

GPunch follows a **client-server architecture** with a cloud-hosted backend and a native Android client.



Backend layers:



Android layers (MVVM):

UI Layer – Activities (Views)
Splash, Login, Register, OTP, Dashboard
ViewModel Layer
AuthViewModel, DashboardViewModel
(LiveData, Coroutines)
Repository Layer
GpunchApiService (Retrofit + OkHttp)
Service / Worker Layer
GeofenceMonitorService (Foreground)
PendingClockOutWorker (WorkManager)
Utility Layer
SessionManager GeofenceUtils

Operating environment:

Component	Specification
Backend Runtime	Node.js >= 18.0.0
Backend Framework	Express.js ^4.18
Database	MongoDB Atlas (M0 free tier or higher)
Deployment	Render, Railway, or any HTTPS Node.js host
Android Min SDK	API 26 (Android 8.0 Oreo)
Android Target SDK	API 34 (Android 14)
Android Language	Kotlin 1.9.23

2. FUNCTIONAL REQUIREMENTS

FR-01: Domain-Restricted Self-Registration

Description: The system allows employees to self-register using only an institutional email address. Registrations from non-institutional domains are automatically rejected.

Inputs: Full name, institutional email, Android device ID (ANDROID_ID)

Processing: 1. Backend loads GeofenceConfig.allowedDomain from the database. 2. Email domain (email.split('@')[1]) is compared against allowedDomain. 3. On mismatch: HTTP 403 returned; INVALID_DOMAIN audit event created. 4. On match: 6-digit numeric OTP generated, stored with 10-minute expiry, sent to the user's email.

Outputs: OTP email delivered to the user, or domain rejection error

ID	Requirement
FR-01.1	The backend shall compare the email domain against <code>GeofenceConfig.allowedDomain</code> on every registration request.
FR-01.2	If the domain does not match, HTTP 403 shall be returned with message "Email domain is not allowed."
FR-01.3	Every rejected domain attempt shall create an <code>INVALID_DOMAIN</code> entry in <code>AuditLog</code> .
FR-01.4	If the domain matches, a 6-digit OTP shall be generated with a 10-minute expiry and emailed to the user.
FR-01.5	Duplicate email registrations shall return HTTP 409.

Priority: High | **Test Cases:** TC-01, TC-02

FR-02: OTP Verification and Device Binding

Description: Upon entering the received OTP, the system verifies it and permanently binds the user's `ANDROID_ID` to their account.

Inputs: Email, 6-digit OTP, `ANDROID_ID`

Processing: 1. Server finds user by email. 2. OTP compared as strict string equality (prevents type-coercion attacks). 3. Expiry checked against current timestamp. 4. On failure: `OTP_FAILED` logged; HTTP 400 returned. 5. On success: `isVerified` = true, `androidId` saved; JWT issued.

Outputs: Signed JWT token, or error with `OTP_FAILED` audit entry

ID	Requirement
FR-02.1	OTP comparison shall use strict string equality to prevent type-coercion injection.
FR-02.2	Expired OTPs shall return HTTP 400 and log <code>OTP_FAILED</code> .
FR-02.3	On success, <code>androidId</code> shall be permanently stored against the user record.
FR-02.4	A signed JWT (HS256, 7-day expiry) shall be issued on success.
FR-02.5	OTP values shall never appear in any API response or log output.

Priority: High | **Test Case:** TC-03

FR-03: Device Binding Enforcement

Description: A verified user may only log in from the device they originally registered with. Login from a different device is rejected and logged.

Inputs: Email, ANDROID_ID

Processing: 1. Login endpoint compares req.body.androidId against user.androidId using strict equality. 2. On mismatch: HTTP 403 returned; UNAUTHORIZED_DEVICE audit event created. 3. On match: OTP sent to registered email for session establishment.

Outputs: OTP sent (if device matches), or device rejection error with audit log entry

ID	Requirement
FR-03.1	The login endpoint shall compare the submitted androidId against the stored binding.
FR-03.2	A mismatch shall return HTTP 403 with message "Device not recognised."
FR-03.3	Every failed device check shall log an UNAUTHORIZED_DEVICE event.
FR-03.4	An admin may clear a device binding to allow re-binding (see FR-10).

Priority: High | **Test Case:** TC-04

FR-04: GPS Settings Check and Location Fetching

Description: Before clock-in, the Android app verifies that high-accuracy GPS is enabled. If GPS is off, an in-app system dialog guides the user to enable it.

Inputs: User taps "Clock In" button

Processing: 1. LocationSettingsRequest built with PRIORITY_HIGH_ACCURACY. 2. LocationServices.getSettingsClient().checkLocationSettings() called. 3. If settings satisfied: getCurrentLocation() proceeds. 4. If not satisfied: ResolvableApiException caught; system settings dialog launched via IntentSenderRequest.

Outputs: GPS coordinates obtained, or system settings dialog displayed

ID	Requirement
FR-04.1	GPS settings shall be checked before every clock-in attempt.
FR-04.2	If GPS is disabled, the Android system settings dialog shall open in-app (no manual navigation).
FR-04.3	Clock-in shall only proceed after valid GPS coordinates are obtained.

Priority: High | **Test Case:** TC-05

FR-05: Geofence-Based Clock-In

Description: Clock-in is permitted only when the user is physically within the configured geofence radius. The server validates coordinates independently of the client.

Inputs: JWT, latitude, longitude, ANDROID_ID

Processing: 1. Server loads GeofenceConfig (centre lat/lng, radius). 2. haversineDistance(config, request) computed server-side. 3. If distance > allowedRadius: HTTP 403 returned; OUT_OF_BOUNDS logged. 4. If distance <= allowedRadius: AttendanceRecord created with clockInTime, clockInLat, clockInLng.

Outputs: Clock-in record created, or out-of-bounds rejection with distance

ID	Requirement
FR-05.1	The server shall compute Haversine distance independently; client assertions are not trusted.
FR-05.2	If distance > allowedRadius, HTTP 403 shall be returned with the actual and allowed distances.
FR-05.3	A successful clock-in shall create an AttendanceRecord with full location and timestamp data.
FR-05.4	The app shall display the computed distance from the geofence centre on the Dashboard.

Priority: High | **Test Cases:** TC-06, TC-07

FR-06: Mock / Fake GPS Detection

Description: The Android client detects GPS spoofing and denies clock-in. The event is reported to the server as a security audit log entry.

Inputs: Location object from Fused Location Provider

Processing: 1. location.isMock checked (API 18+; reliable from API 31+). 2. Settings.Secure.ALLOW MOCK_LOCATION checked for devices below API 31. 3. If either returns true: clock-in denied; toast displayed; MOCK_LOCATION reported to POST /api/audit.

Outputs: Clock-in denied with user-visible message; MOCK_LOCATION audit entry created

ID	Requirement
FR-06.1	location.isMock shall be evaluated on every location fix before clock-in.
FR-06.2	The legacy ALLOW MOCK_LOCATION setting shall additionally be checked for pre-API-31 devices.
FR-06.3	If mock location is detected, clock-in shall be denied and a toast message displayed.
FR-06.4	The app shall report a MOCK_LOCATION event to POST /api/audit.

Priority: High | **Test Case:** TC-08

FR-07: Background Geofence Monitoring with Persistent Notification

Description: After a successful clock-in, a foreground service monitors the user's GPS position continuously and displays a persistent notification.

Inputs: Successful clock-in event

Processing: 1. GeofenceMonitorService started as an Android Foreground Service. 2. Persistent notification shown in the Android notification tray. 3. GPS polled via FusedLocationProviderClient.requestLocationUpdates() with adaptive intervals: 15 s when moving (≥ 5 m displacement), 60 s when stationary.

Outputs: Persistent notification visible in tray; continuous location monitoring active

ID	Requirement
FR-07.1	GeofenceMonitorService shall be started as a Foreground Service immediately after successful clock-in.
FR-07.2	A persistent notification shall appear in the Android tray during the active session.
FR-07.3	Adaptive GPS polling (15 s / 60 s) shall be used to minimise battery drain.
FR-07.4	The service shall survive app backgrounding, task kill, and screen-off events.

Priority: High | **Test Case:** TC-09

FR-08: Automatic Clock-Out on Geofence Breach

Description: When the monitoring service detects that the user has left the geofence, it automatically calls the clock-out API and notifies the user.

Inputs: GPS coordinates from GeofenceMonitorService indicating boundary exit

Processing: 1. haversineDistance computed on each location update. 2. If distance > fenceRadius: POST /api/punch/out called with autoClockOut: true. 3. SessionManager.setIsClockedIn(false) updated. 4. Push notification displayed: "You have left the work zone. Attendance has been recorded." 5. Foreground service stops.

Outputs: Clock-out record persisted; user notified; background service stopped

ID	Requirement
FR-08.1	A geofence exit shall trigger POST /api/punch/out with autoClockOut: true automatically.
FR-08.2	The app session state shall be updated to "clocked out" immediately.
FR-08.3	A push notification shall inform the user that auto clock-out has occurred.
FR-08.4	The foreground service shall stop after auto clock-out completes.

Priority: High | **Test Case:** TC-10

FR-09: Admin – Geofence and Domain Configuration

Description: The system administrator sets the geofence centre, radius, and allowed email domain. Changes take effect immediately for all subsequent clock-in attempts.

Inputs: JWT (admin role), latitude, longitude, radius (metres), email domain

Endpoint: POST /api/admin/geofence

Processing: Upserts a single GeofenceConfig document. All subsequent /api/punch/in calls load this configuration.

Outputs: Updated GeofenceConfig confirmed

ID	Requirement
FR-09.1	The endpoint shall accept { latitude, longitude, allowedRadius, allowedDomain } with admin JWT.
FR-09.2	The operation shall upsert, ensuring only one GeofenceConfig document exists at any time.
FR-09.3	Changes shall take effect for all clock-in requests that arrive after the upsert.

Priority: High | **Test Case:** TC-11

FR-10: Admin — Device Unlinking

Description: The administrator can clear a user's device binding, allowing them to re-bind from a new device on the next OTP verification.

Inputs: JWT (admin role), user ID

Endpoint: POST /api/admin/reset-device/:userId

Processing: Sets user.androidId = null. On the user's next OTP verification, the new device ANDROID_ID is bound.

ID	Requirement
FR-10.1	The endpoint shall set user.androidId = null for the specified user.
FR-10.2	After the reset, the user shall be able to bind a new device on next successful OTP verification.

Priority: High | **Test Case:** TC-12

FR-11: Admin — Security Audit Log Viewing

Description: The administrator can view a paginated, filterable list of all security events to monitor for attendance fraud attempts.

Inputs: JWT (admin role), optional eventType filter, page, limit

Endpoint: GET /api/admin/audit-logs

Logged Event Types

Event Type	Trigger
INVALID_DOMAIN	Registration with non-institutional email (TC-01)
OTP_FAILED	Incorrect or expired OTP submitted (TC-03)
UNAUTHORIZED_DEVICE	Login from an unbound device (TC-04)
MOCK_LOCATION	Fake GPS app detected on client (TC-08)
OUT_OF_BOUNDS	Clock-in attempted outside the geofence
SUSPICIOUS_ACTIVITY	Other anomalies reported by the client

ID	Requirement
FR-11.1	The endpoint shall support eventType, page, and limit query parameters.
FR-11.2	The eventType filter shall be validated against a whitelist to prevent injection.
FR-11.3	Each log entry shall include: userId, email, androidId, eventType, timestamp, ipAddress, metadata.

Priority: High | **Test Case:** TC-13

FR-12: Offline Clock-Out Queueing

Description: If the auto clock-out API call fails due to no network connectivity, the payload is persisted locally by WorkManager and retried automatically when the network is restored.

Inputs: Network failure during auto clock-out

Processing: 1. IOException (or any exception) caught in GeofenceMonitorService. 2. PendingClockOutWorker enqueued via WorkManager with a CONNECTED constraint. 3. SessionManager.setIsClockedin(false) set immediately (optimistic update). 4. When connectivity returns, WorkManager fires the worker which calls POST /api/punch/out.

Outputs: Clock-out record persisted once network is available; no data lost

ID	Requirement
FR-12.1	Any network failure during auto clock-out shall enqueue a PendingClockOutWorker via WorkManager.
FR-12.2	The WorkManager job shall have a CONNECTED network constraint.
FR-12.3	The clock-out payload shall persist across app restarts and device reboots.
FR-12.4	The app session state shall be updated to "clocked out" optimistically before retry.

Priority: Medium | **Test Case:** TC-14

FR-13: Admin — CSV Attendance Export

Description: The administrator can download a complete attendance report for all users in CSV format.

Endpoint: GET /api/admin/export-csv (admin JWT required)

Output columns: Name, Email, Clock-In Time, Clock-Out Time, Duration (minutes), Auto Clock-Out flag, Android ID

ID	Requirement
FR-13.1	The endpoint shall return a downloadable CSV file with a Content-Disposition: attachment header.
FR-13.2	Every AttendanceRecord document shall appear as one row in the export.

Priority: Medium

Test Case Traceability Matrix

TC	Requirement	Input / Action	Expected Output	FR	Result
TC-01	Domain Restriction - Reject	Register with user@gmail.com	HTTP 403; IN-VALID_DOMAIN logged	FR-01	Pass
TC-02	Domain Restriction - Accept	Register with user@psgtech.ac.in	HTTP 200; OTP sent	FR-01	Pass
TC-03	OTP Verification Failure	Submit wrong 6-digit OTP	HTTP 400; OTP_FAILED logged	FR-02	Pass
TC-04	Device Binding Enforcement	Login from Device 2 (bound to Device 1)	HTTP 403; UNAUTHORIZED_DEVICE logged	FR-03	Pass
TC-05	GPS Settings Check	Tap Clock-In with GPS disabled	System location settings dialog shown	FR-04	Pass
TC-06	Geofence Inside	Clock-in at 20 m from fence centre	HTTP 201; foreground service starts	FR-05	Pass
TC-07	Geofence Outside	Clock-in at 200 m from fence centre	HTTP 403; distance in response	FR-05	Pass
TC-08	Mock Location	Enable Fake GPS app; attempt clock-in	Denied; toast shown; MOCK_LOCATION logged	FR-06	Pass
TC-09	Background Tracking	Minimise app after Clock-In	Persistent notification visible in tray	FR-07	Pass
TC-10	Auto Clock-Out	Move 100+ m from campus boundary	Clock-out API called; user notified	FR-08	Pass

TC	Requirement	Input / Action	Expected Output	FR	Result
TC-11	Admin Geofence Config	Change radius 50 m -> 10 m	Users at 20 m rejected on next clock-in	FR-09	Pass
TC-12	Admin Device Reset	Admin resets binding for User A	User A can bind new device on next login	FR-10	Pass
TC-13	Audit Log Visibility	Admin views security dashboard	TC-01, TC-03, TC-04, TC-08 events visible	FR-11	Pass
TC-14	Offline Clock-Out Queue	Auto clock-out in Airplane Mode	Payload queued; sent on reconnect	FR-12	Pass

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 User Interface Requirements

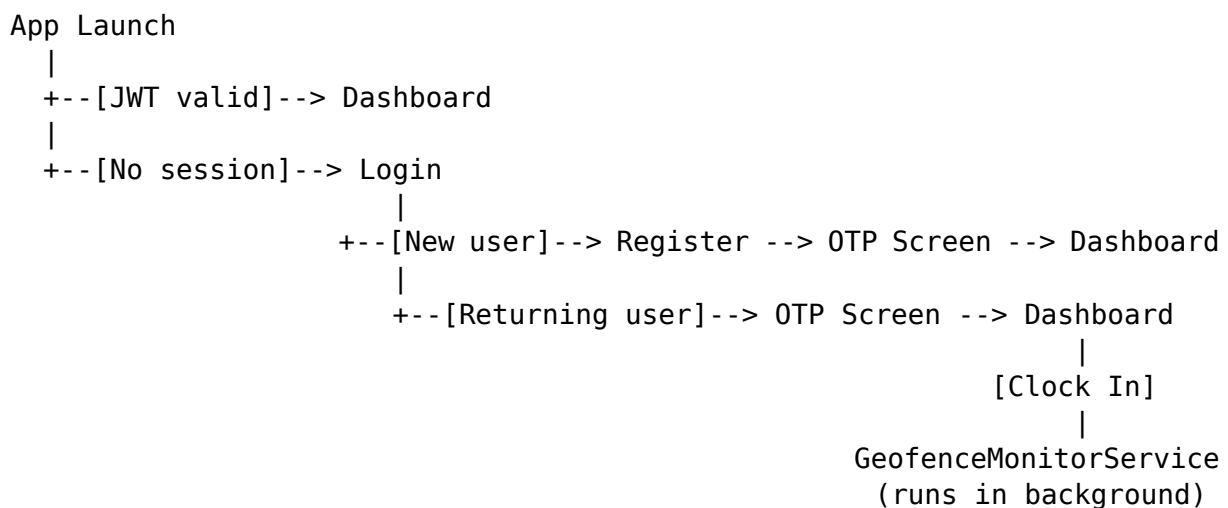
Android Application Screens

Screen	Key Elements	Description
Splash	Logo, progress bar	Auto-redirects: Dashboard (if JWT valid) or Login
Login	Email input, "Send OTP" button	Passwordless login via institutional email
Register	Name + email inputs, "Send OTP" button	New account with domain validation
OTP Verification	6-digit code input, countdown timer (60 s), Resend link	OTP entry with resend throttle
Dashboard	Clock-In/Out button, distance indicator, session timer, status badge, History, Logout	Primary user interface for all attendance actions

UI Behaviour Requirements

ID	Requirement
UI-01	All screens shall display human-readable error messages without technical stack traces.
UI-02	The Dashboard shall display the current clock-in status ("Clocked In" / "Clocked Out") as a prominent status badge.
UI-03	The OTP screen shall show a 60-second countdown before the Resend button becomes active.
UI-04	The persistent notification shall contain the session start time and one-tap navigation to the Dashboard.
UI-05	The Clock-In button shall be disabled while a GPS fix or network request is in progress.

Authentication and Navigation Flow



3.2 Hardware Interface Requirements

Hardware	Requirement
GPS Receiver	Required for coordinate acquisition; high-accuracy (GNSS) mode preferred
Network Interface	Wi-Fi or mobile data (2G minimum) for REST API communication
Notification System	Android notification tray for persistent session notification and auto clock-out alert
Internal Storage	SharedPreferences for JWT token, session state, and WorkManager job persistence
Clock / RTC	Device system time used for OTP expiry checks and attendance timestamps

3.3 Software Interface Requirements

Interface	Technology	Version	Purpose
MongoDB Atlas	Mongoose ODM	^8.3	Persistent storage for all four collections
Gmail SMTP	Nodemailer	^8.0	Transactional OTP email delivery
Fused Location Provider	Google Play Services	21.3.0	High-accuracy GPS coordinate acquisition
WorkManager	AndroidX WorkManager	2.9.0	Persistent background job scheduling
Retrofit + OkHttp	Retrofit / OkHttp	2.11 / 4.12	Android HTTP client with JWT interceptor
express-validator	npm	^7.0	Server-side request body validation
helmet.js	npm	^7.1	HTTP security header hardening
express-rate-limit	npm	^7.2	IP-based request rate limiting
jsonwebtoken	npm	^9.0	JWT signing and verification (HS256)

3.4 Communication Interface Requirements

Protocol	Usage	Details
HTTPS (TLS 1.2+)	All client-server API communication	Enforced by OkHttp CertificatePinner on Android; mandatory on Render/Railway deployments
JSON	Request and response body format	Content-Type: application/json on all endpoints
Bearer JWT	API session authentication	Authorization: Bearer <token> header on all protected routes
SMTP / STARTTLS	OTP email delivery	Gmail port 587 with STARTTLS; App Password authentication

Rate Limiting

Scope	Limit
Auth endpoints (/api/auth/*)	20 requests per 15 minutes per IP
Global (all routes)	100 requests per 15 minutes per IP

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Security

ID	Requirement
NFR-S-01	All API endpoints except /api/auth/* and /health shall require a valid JWT (HS256).
NFR-S-02	Admin routes shall additionally verify role: "admin" in the JWT payload.
NFR-S-03	OTP values shall never appear in API responses, server logs, or database query results (stripped via Mongoose toJSON transform).
NFR-S-04	All Mongoose queries shall use explicit type casting to prevent NoSQL operator injection attacks.
NFR-S-05	The eventType filter parameter on audit-log queries shall be validated against a whitelist.
NFR-S-06	All HTTP response headers shall be hardened using helmet.js (X-Frame-Options, CSP, HSTS, etc.).
NFR-S-07	Request body size shall be capped at 10 KB to mitigate payload-flooding attacks.
NFR-S-08	HTTPS shall be enforced on all production deployments; HTTP connections shall be rejected or redirected.
NFR-S-09	JWT tokens shall expire after 7 days; OTPs shall expire after 10 minutes.
NFR-S-10	GPS coordinates shall be validated server-side; client-provided coordinates shall never be trusted without re-validation.

Security Threat Model

Threat	Mitigation
Buddy punching	Device binding: each account locked to one ANDROID_ID

Threat	Mitigation
GPS spoofing	location.isMock check + legacy developer-options check + server-side Haversine
Proxy API calls	Server-side Haversine validation; client cannot override distance result
NoSQL injection	Explicit typed Mongoose queries; eventType whitelisting
JWT forgery	HS256 with long JWT_SECRET; 7-day expiry
OTP brute-force	Auth rate limit: 20 req/15 min; 10-minute OTP expiry
Replay attacks	OTP is single-use (nulled after verification); JWT expiry

4.2 Capacity

ID	Requirement
NFR-C-01	The system shall support at least 500 registered users on a MongoDB Atlas M0 (free tier) single-campus deployment.
NFR-C-02	The backend shall handle at least 50 concurrent clock-in requests without degradation.
NFR-C-03	AttendanceRecord collection shall support at least 2 years of daily records for 500 users without requiring index changes.
NFR-C-04	AuditLog collection shall be indexed on { eventType, createdAt } and { userId, createdAt } to support efficient admin queries.

4.3 Compatibility

ID	Requirement
NFR-CP-01	The Android client shall run on Android 8.0 (API 26) and all higher versions up to Android 14 (API 34).
NFR-CP-02	The Android client shall function on ARM, ARM64, and x86_64 processor architectures.
NFR-CP-03	The backend shall run on any Node.js >= 18.0.0 compatible hosting platform.

ID	Requirement
NFR-CP-04	The backend API shall be accessible from any HTTP client that supports HTTPS and JSON (Retrofit, Postman, curl).
NFR-CP-05	Google Play Services shall be available on the target Android device (required for FLP and WorkManager).

4.4 Reliability

ID	Requirement
NFR-R-01	Offline clock-out payloads shall not be lost; WorkManager persistence survives app restarts and device reboots.
NFR-R-02	BootReceiver shall restore the GeofenceMonitorService foreground service after a device reboot if the user was clocked in at shutdown.
NFR-R-03	The backend deployment on Render/Railway shall achieve $\geq 99\%$ uptime during normal workday hours (08:00-18:00).
NFR-R-04	Failed WorkManager jobs shall be retried with exponential back-off until the CONNECTED constraint is satisfied.

4.5 Scalability

ID	Requirement
NFR-SC-01	The backend architecture shall allow horizontal scaling by adding additional Node.js instances behind a load balancer without schema changes.
NFR-SC-02	MongoDB Atlas connection pooling (via Mongoose) shall allow the system to scale to Atlas M10 or higher tiers without code changes.
NFR-SC-03	The system design shall allow the addition of a second geofence (multi-campus) in a future release with only additive schema changes.
NFR-SC-04	Rate limiting configuration shall be externalisable to environment variables to support traffic growth without code redeployment.

4.6 Maintainability

ID	Requirement
NFR-M-01	Backend code shall follow MVC separation: models/, routes/ (controllers), middleware/, utils/.
NFR-M-02	Android code shall follow MVVM: Activities (Views) observe LiveData from ViewModels; network calls are made through GpunchApiService.
NFR-M-03	All secrets and configuration values shall be externalised to .env files; no hardcoded secrets in source code.
NFR-M-04	Backend unit tests shall cover $\geq 80\%$ of utility functions (Haversine, OTP) and critical auth flows.
NFR-M-05	The .env.example file shall document all required environment variables with descriptions.

4.7 Usability

ID	Requirement
NFR-U-01	Registration shall require no passwords — only name, institutional email, and a 6-digit OTP.
NFR-U-02	The OTP resend timer (60 seconds) shall prevent accidental double-sends while allowing genuine resends.
NFR-U-03	All user-facing error messages shall be human-readable; no technical stack traces shall be displayed to end users.
NFR-U-04	The persistent foreground notification shall provide one-tap navigation back to the Dashboard activity.
NFR-U-05	The Clock-In button shall be visually disabled (greyed out) when the user is already clocked in.
NFR-U-06	The app shall display the distance from the geofence centre on the Dashboard so the user can understand rejection reasons.

4.8 Other Non-Functional Requirements

Legal and Compliance

ID	Requirement
NFR-O-01	Location data (GPS coordinates) shall be stored only for legitimate attendance tracking purposes and shall not be shared with third parties.
NFR-O-02	No biometric data (fingerprint, face, voice) shall be collected or stored by the system.
NFR-O-03	ANDROID_ID is a device-level pseudonymous identifier; it does not directly identify a natural person.
NFR-O-04	Institutional administrators are responsible for ensuring the system is deployed in compliance with applicable data protection regulations (e.g., DPDP Act 2023).

Performance

ID	Requirement
NFR-O-05	Clock-in API response time shall be $\leq$ 2 seconds under normal network conditions.
NFR-O-06	The app shall obtain a GPS fix within 10 seconds of the Dashboard opening.
NFR-O-07	Background service battery impact shall be minimised through adaptive GPS polling (15 s active / 60 s stationary).

5. DEFINITIONS AND ACRONYMS

5.1 Acronyms

Acronym	Full Form
API	Application Programming Interface
ANDROID_ID	A 64-bit hexadecimal identifier unique to an Android device per app install and signing key
CSV	Comma-Separated Values
FLP	Fused Location Provider (Google Play Services location API)
FR	Functional Requirement
GPS	Global Positioning System
GNSS	Global Navigation Satellite System
HTTP / HTTPS	HyperText Transfer Protocol / Secure

Acronym	Full Form
JWT	JSON Web Token
MCA	Master of Computer Applications
MVVM	Model-View-ViewModel
NFR	Non-Functional Requirement
ODM	Object Document Mapper
OTP	One-Time Password
REST	Representational State Transfer
SDK	Software Development Kit
SMTP	Simple Mail Transfer Protocol
SRS	Software Requirements Specification
TC	Test Case
TLS	Transport Layer Security

5.2 Definitions

Term	Definition
Attendance Record	A database document storing a single clock-in / clock-out session for one user, including timestamps, GPS coordinates, duration, and device ID.
Audit Log	A tamper-evident security event record capturing the type, actor, device, location, and timestamp of a security-relevant action (e.g., domain rejection, OTP failure, device mismatch).
Buddy Punching	An attendance fraud where one employee clocks in or out on behalf of another absent employee.
Device Binding	The permanent association of a user account with a specific ANDROID_ID established during first OTP verification.
Foreground Service	An Android service that runs while displaying a persistent notification, making it resistant to system termination.
Geofence	A virtual geographic boundary defined by a centre latitude/longitude coordinate pair and a radius in metres.
Haversine Formula	A mathematical formula that computes the great-circle distance between two points on a sphere given their latitude and longitude.
JWT (JSON Web Token)	A compact, self-contained token that encodes user identity and role claims, signed with HMAC-SHA256, used for stateless API session management.

Term	Definition
Mock Location	A falsified GPS location generated by a third-party application for the purpose of deceiving location-based systems.
OTP (One-Time Password)	A 6-digit numeric code valid for 10 minutes, generated server-side and delivered by email, used in place of a password for authentication.
Proxy Attendance	An attendance fraud where a person submits an API request on behalf of another person who is not physically present.
WorkManager	An AndroidX library that schedules deferrable, constraint-based background tasks that persist across app restarts and device reboots.

5.3 Data Model Summary

Collection	Primary Fields	Purpose
users	name, email, role, androidId, otp, otpExpiresAt, isVerified, isActive	User accounts and device binding
attendancerecords	userId, email, clockInTime, clockOutTime, durationMinutes, autoClockOut, clockInLat/Lng	Attendance history
geofenceconfigs	latitude, longitude, allowedRadius, allowedDomain, updatedBy	Active geofence parameters (singleton)
auditlogs	userId, email, androidId, eventType, ipAddress, latitude, longitude, metadata	Security event audit trail

5.4 Revision History

Version	Date	Author	Change Description
0.1	April 2026	Kavin M	Initial draft — scope and FR outline

Version	Date	Author	Change Description
0.9	April 2026	Shanmugappriya K	NFRs, data models, API specification, test matrix
1.0	May 2026	Kavin M, Shanmugappriya K	Final — all 14 test cases verified; 23MX21 template applied

Document ends — GPunch SRS v1.0 (Template: 23MX21)